Explain one cycle of the boat engine

Task 3.5 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Draw a four-frame loop. Label the chamber, heat, water and tube mouths. Distinguish inferred pressure change from visible motion. Show how cooler fluid can enter again.

Mark heat input, surrounding cooler water, and tube openings. Identify where the team observed water motion.

State	Heat transfer	Pressure change inferred	Flow direction
Warm-up			
Expansion			
Cooling			
Return			

02 Plan it on paper

Draw successive states of the working fluid. Predict the direction of flow at two points in the cycle.

My prediction, chosen feature or design limit

Explain one cycle of the boat engine

Task 3.5 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Watch the approved demonstration. Annotate the cycle with changes in temperature, pressure, and movement. Discuss which changes are inferred. Have a partner trace the entire cycle back to its starting state. Repair a missing energy transfer or reversed arrow.

Claim	Observation	Inference or open question
Water motion		
Repeating cycle		

04 Use the reading

Ch. 20 | An Ingenious Engine | printed pp. 185-190

Ch. 21 | Energy from the Earth | printed pp. 191-198

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Use two stages in the cycle to explain how heating can produce repeated motion. Identify evidence and one inference.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.